Conical F-PSM closed forms: comparison with vector rigorous coupled-wave analysis and validated diffraction-efficiency bounds

PSM PROJECT TECHNICAL REPORT

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Conical phase-grating closed forms are derived from locally refracted parallel-stacked-mirror (PSM) equations in their exact Bremmer representation. Floquet expansion and a change to surface-directed electromagnetic amplitudes give explicit vector coupling vertices. Perturbative elimination of nonprincipal waves yields cubic bulk and quadratic face corrections. An independent implementation of Moharam–Gaylord vector rigorous coupled-wave analysis supplies a 504-point comparison, with improvement at every sampled case. Complex ball arithmetic certifies 24 finite-system comparisons. A separate volume-integral residual argument bounds the infinite Fourier tail at 27 thinner-grating points, giving full Maxwell efficiency-error bounds below 6.4×10^{-7}. A full Maxwell certificate is not established for the stronger Bragg-conversion sweep.

1. INTRODUCTION

Moharam and Gaylord formulated rigorous three-dimensional vector coupled-wave analysis for planar gratings with arbitrary incidence direction and slanted fringes [1]. Their formulation retains coupling between both polarisations and both propagation directions. It therefore supplies an appropriate published reference for conical replay of a volume phase grating. Later work addressed stable numerical implementation and the difficulties associated with evanescent waves [2, 3].

The conical F-PSM model considered here retains a four-amplitude principal system, cubic bulk coefficients and quadratic face corrections. Section 3 derives these from the PSM/Bremmer starting point, distinguishing exact changes of representation from the subsequent finite-order approximation. Its thickness dependence has an analytic matrix-function representation, reducible to quartic radicals. Agreement between its radical evaluation and another evaluation of the same four-dimensional system checks algebra and numerical evaluation, but does not establish accuracy against the full grating problem.

This report makes three distinct comparisons. First, an independently assembled Moharam–Gaylord RCWA calculation provides a numerical reference over a declared parameter grid. Second, complex ball arithmetic encloses both the closed-form and finite-RCWA results, yielding rigorous bounds on their difference. Third, for a smaller parameter set, a volume-integral residual estimate also bounds the infinite diffraction-order tail. The third result certifies an error relative to the ideal full Maxwell problem; the second alone does not.

The RCWA code is a new implementation of the published equations, not software supplied by Moharam or Gaylord. The original paper was inspected directly, including its Eqs. (20)–(24), boundary conditions and efficiency definitions. Its reprint is available in a Georgia Tech institutional report, pages 62–69 of the archived PDF [4]. No claim of reproducing unpublished original numerical data is made.

2. GEOMETRY AND REFERENCE FORMULATION

A. Material and coordinate conventions

The slab occupies $0 < y < d$ in the F-PSM coordinates (x', y, z), with

$$n(\xi) = n_0 + n_1 \cos(K\xi + \phi_g), \tag{1}$$

$$\xi = -x' \sin\Psi + y \cos\Psi, \quad K = 2\pi/\Lambda. \tag{2}$$

The indices are real, the medium is nonmagnetic, and both exterior indices are n_0. Define $k_0 = 2\pi/\lambda$, $\beta = k_0 n_0$, $\mu = n_1/n_0$, $K_x = K \sin\Psi$ and $K_y = K \cos\Psi$. The time convention is $e^{+i\omega t}$.

Replay has polar angle θ from the surface normal and azimuth φ from $+x'$ toward $+z$. Thus altitude above the plate is $a = \pi/2 - \theta$, and

$$\alpha = \beta \sin\theta \cos\varphi, \qquad q = \beta \sin\theta \sin\varphi, \tag{3}$$

$$\alpha_m = \alpha + mK_x, \qquad \gamma_m = \sqrt{\beta^2 - \alpha_m^2 - q^2}. \tag{4}$$

Use $\gamma_m > 0$ for an open order and $\gamma_m = -i\kappa_m$, $\kappa_m > 0$, for a closed order. The principal output is order 1: its direction is $\tau = +1$ for transmission and $\tau = -1$ for reflection. Grazing

orders require a separate limiting treatment and are not certified here.

For direct correspondence with Ref. [1], use the proper rotation

$$(X, Y, Z) = (-x', z, y). \tag{5}$$

The paper's normal is Z, its grating vector is $(K_x, 0, K_y)$ and its tangential order wave number is $a_m = -\alpha_m$. The paper's incidence-plane angle is $\delta = \pi - \varphi$ for the geometries used here. The two tangential electric and magnetic fields form

$$f = (E_X, E_Y, Z_0 H_X, Z_0 H_Y)^{\mathsf{T}}, \qquad f' = -\mathrm{i} G_N f, \tag{6}$$

after removing the order-dependent phase $e^{\mathrm{i}mK_y Z}$. Z_0 denotes the vacuum impedance; a prime denotes differentiation with respect to slab depth.

B. Exact material Fourier coefficients

For $N = 2M + 1$ retained orders $m = -M, \ldots, M$, define

$$\mathcal{E}_{mn} = \varepsilon_{m-n}, \qquad \mathcal{A}_{mn} = a_{m-n}, \tag{7}$$

where ε_h and a_h are the Fourier coefficients of n^2 and n^{-2}, respectively. For Eq. (1),

$$\varepsilon_h = e^{\mathrm{i}h\phi_g} \begin{cases} n_0^2 + n_1^2/2, & h = 0, \\ n_0 n_1, & |h| = 1, \\ n_1^2/4, & |h| = 2, \\ 0, & |h| > 2. \end{cases} \tag{8}$$

Writing $\Delta = \sqrt{n_0^2 - n_1^2}$ and $r = n_1/(n_0 + \Delta)$ gives

$$a_h = e^{\mathrm{i}h\phi_g} (-r)^{|h|} \frac{n_0 + |h|\Delta}{\Delta^3}. \tag{9}$$

Equation (9) follows by differentiating the Fourier series of $(n_0 + n_1 \cos\chi)^{-1}$ with respect to n_0, holding n_1 fixed. It was checked independently by quadrature. The DC and second-harmonic terms in Eq. (8) are retained; replacing a sinusoidal index by a sinusoidal permittivity changes the model at second order.

C. Moharam–Gaylord system and finite factorisation

Let $A = \mathrm{diag}(a_m)$ and $L = \mathrm{diag}(mK_y)$. Equations (20)–(23) of Ref. [1] give

$$G_N = \begin{pmatrix} L & 0 & P_{11} & P_{12} \\ 0 & L & P_{21} & P_{22} \\ Q_{11} & Q_{12} & L & 0 \\ Q_{21} & Q_{22} & 0 & L \end{pmatrix}, \tag{10}$$

with

$$\begin{aligned} P_{11} &= qA\mathcal{A}/k_0, & P_{12} &= k_0 \mathsf{I} - A\mathcal{A}A/k_0, \\ P_{21} &= q^2\mathcal{A}/k_0 - k_0\mathsf{I}, & P_{22} &= -q\mathcal{A}A/k_0, \\ Q_{11} &= -qA/k_0, & Q_{12} &= A^2/k_0 - k_0\mathcal{E}, \\ Q_{21} &= k_0\mathcal{E} - q^2\mathsf{I}/k_0, & Q_{22} &= qA/k_0. \end{aligned} \tag{11}$$

The finite Moharam system uses the Toeplitz matrix of the exact reciprocal coefficients, $\mathcal{A} = T_N(1/\varepsilon)$. This differs at finite N from $T_N(\varepsilon)^{-1}$, used in the earlier project reference. Both choices are examined. The latter is denoted the electric-field Galerkin system below and is used in the infinite-tail proof because it is the exact projection of the electric volume-integral equation.

For numerical propagation, a short constant-coefficient step is evaluated by a matrix exponential and converted to a scattering matrix. Repeated scattering composition gives the requested thickness. The short steps do not approximate a depth-varying material: the generator in Eq. (6) is constant. The operation is algebraically identical to full propagation and boundary matching, and avoids directly evaluating growing evanescent factors across the entire slab. Appendix A gives the composition explicitly.

D. Flux and polarisation conventions

For each open order, s is perpendicular to the plane containing that order's wavevector and the surface normal; $p = s \times \widehat{k}$. A single common incident s, p basis is used in both calculations. If $t_{u\ell}$ and $r_{u\ell}$ are unit-electric-field amplitudes, the order-1 flux Jones matrices are

$$J_T = \sqrt{\gamma_1/\gamma_0}\, t_1, \qquad J_R = \sqrt{\gamma_1/\gamma_0}\, r_1. \tag{12}$$

For a normalised complex incident vector $b \in \mathbb{C}^2$, the total order-1 efficiency is $\eta(b) = \|Jb\|_2^2$. This definition includes coherent and elliptical polarisations. A power sum over all propagating orders is also checked, but power conservation alone is not a truncation-error bound. Moharam and Gaylord explicitly make this distinction in their discussion following Eq. (43) [1].

3. DERIVATION FROM THE PSM–BREMMER SYSTEM

A. Locally refracted PSM in a conical momentum sector

The PSM construction resolves a continuous index profile into infinitesimal, fringe-parallel interfaces and combines their Fresnel reflections [5]. The starting point here is its *locally refracted* directional form, equivalently the exact Bremmer representation of the scalar field equations [6]. Holding a ray angle fixed through the modulated medium would already be an approximation; that is not the starting point used below. The relation of Bremmer formulations to rigorous grating scattering has also been studied independently [7].

Introduce the fringe-parallel coordinate and unit vectors

$$\begin{aligned} &\tau_f = x' \cos\Psi + y \sin\Psi, \\ &\boldsymbol{e}_t = (\cos\Psi, \sin\Psi, 0), \qquad \boldsymbol{e}_\xi = (-\sin\Psi, \cos\Psi, 0). \end{aligned} \tag{13}$$

For one bulk momentum sector, suppress the factor $e^{-\mathrm{i}(p\tau_f + qz)}$ and define

$$\rho^2 = p^2 + q^2, \qquad k^2(\xi) = k_0^2 \varepsilon(\xi) - \rho^2. \tag{14}$$

Both p and q are conserved within this sector. The conical extension at this stage is precisely $p^2 \mapsto p^2 + q^2$. These are algebraic squares, including for complex Bloch modes. Define

$$\boldsymbol{t} = (p\boldsymbol{e}_t + q\boldsymbol{e}_z)/\rho, \qquad \boldsymbol{s}_f = \boldsymbol{e}_\xi \times \boldsymbol{t}. \tag{15}$$

The removable $\rho = 0$ basis singularity is handled by continuity. With $\boldsymbol{g} = Z_0 \boldsymbol{H}$, the fringe TE and TM families, denoted σ_f and π_f, are

$$\begin{aligned} \sigma_f: &\quad \boldsymbol{E} = \boldsymbol{s}_f u, && u'' + k^2 u = 0, \\ \pi_f: &\quad \boldsymbol{g} = \boldsymbol{s}_f v, && v'' - \frac{\varepsilon'}{\varepsilon} v' + k^2 v = 0. \end{aligned} \tag{16}$$

Primes in this subsection mean differentiation with respect to ξ. The curl equations reconstruct all remaining components:

$$\sigma_f: \quad \boldsymbol{g} = \frac{\rho}{k_0}\boldsymbol{e}_{\tilde{\xi}} u - \frac{\mathrm{i}}{k_0}\boldsymbol{t}\, u',$$
$$\pi_f: \quad \boldsymbol{E} = -\frac{\rho}{k_0\varepsilon}\boldsymbol{e}_{\tilde{\xi}} v + \frac{\mathrm{i}}{k_0\varepsilon}\boldsymbol{t}\, v'. \tag{17}$$

These are fringe-based families, not the exterior s, p polarisations.

For either scalar $f = u$ or v, adopt the original PSM directional convention

$$f = R - S, \qquad f' = -\mathrm{i}k(R + S). \tag{18}$$

The infinitesimal-interface coefficients are half the logarithmic derivatives of the TE admittance k/k_0 and the TM magnetic-field impedance $k/(k_0\varepsilon)$. Thus

$$\zeta_{\sigma_f} = \frac{k'}{2k}, \qquad \zeta_{\pi_f} = \frac{k'}{2k} - \frac{\varepsilon'}{2\varepsilon}. \tag{19}$$

In the spatial PSM form, let $\sin\vartheta = \rho/(k_0 n)$, $\cos\vartheta = k/(k_0 n)$ and $\partial_t = \boldsymbol{t}\cdot\nabla_{\parallel}$. The locally refracted directional equations are

$$(\sin\vartheta\,\partial_t + \cos\vartheta\,\partial_{\tilde{\xi}})R = -\mathrm{i}k_0 nR - \cos\vartheta\,\zeta_j(R+S),$$
$$(\sin\vartheta\,\partial_t - \cos\vartheta\,\partial_{\tilde{\xi}})S = -\mathrm{i}k_0 nS + \cos\vartheta\,\zeta_j(R+S). \tag{20}$$

For TE, $\cos\vartheta\,\zeta_{\sigma_f} = n'/(2n\cos\vartheta)$, the refracted PSM mirror coefficient. For TM there is the additional $-(n'/n)\cos\vartheta$ term. Substitution of $\partial_t = -\mathrm{i}\rho$ uses $(k_0 n - \rho\sin\vartheta)/\cos\vartheta = k$ and gives, for $j = \sigma_f, \pi_f$,

$$\boxed{\begin{aligned} R' &= -\mathrm{i}kR - \zeta_j(R+S), \\ S' &= +\mathrm{i}kS - \zeta_j(R+S). \end{aligned}} \tag{21}$$

For example, differentiating Eq. (18) using Eq. (21) gives

$$f'' = -k^2 f + \left(\frac{k'}{k} - 2\zeta_j\right) f', \tag{22}$$

which is exactly Eq. (16) for each choice in Eq. (19). Conversely,

$$R = \tfrac{1}{2}(f + \mathrm{i}f'/k), \qquad S = \tfrac{1}{2}(\mathrm{i}f'/k - f) \tag{23}$$

recovers the directional system wherever $k \neq 0$. This establishes the exact starting equivalence. In a uniform medium the negative-going scalar amplitude is $-S$, a sign that must not be lost in subsequent boundary matching. A zero of k makes these directional coordinates singular, although the underlying scalar field equations remain valid.

To exhibit the Bremmer reflection series explicitly, put $Y_{\sigma_f} = k/k_0$, $Y_{\pi_f} = k/(k_0\varepsilon)$, $\widetilde{R} = \sqrt{Y_j}R$, $\widetilde{S} = \sqrt{Y_j}S$ and $\Phi' = k$. Then $A = e^{\mathrm{i}\Phi}\widetilde{R}$, $B = e^{-\mathrm{i}\Phi}\widetilde{S}$ satisfy

$$A' = -\zeta_j e^{2\mathrm{i}\Phi} B, \qquad B' = -\zeta_j e^{-2\mathrm{i}\Phi} A. \tag{24}$$

Integrating these equations with opposite-face incoming data and iterating generates the multiple-reflection series. We retain the differential system before making a Fourier-order approximation.

B. Floquet expansion of the original directional amplitudes

For either family write

$$\begin{pmatrix} R \\ S \end{pmatrix} = e^{-\mathrm{i}p\tau_f - \mathrm{i}qz - \mathrm{i}\nu\xi} \sum_{m\in\mathbb{Z}} \begin{pmatrix} R_m \\ S_m \end{pmatrix} e^{\mathrm{i}mK\xi}. \tag{25}$$

Here ν is a fringe-normal Bloch exponent, distinct from the modulation μ. Let k_h and $\zeta_{j,h}$ be the Fourier coefficients of k and ζ_j at fixed (p, q). Direct substitution into Eq. (21) gives

$$(\nu - mK)R_m = \sum_h \{k_h R_{m-h} - \mathrm{i}\zeta_{j,h}(R_{m-h} + S_{m-h})\},$$
$$(\nu - mK)S_m = \sum_h \{-k_h S_{m-h} - \mathrm{i}\zeta_{j,h}(R_{m-h} + S_{m-h})\}. \tag{26}$$

The sums are infinite. A sinusoidal index does not make k or ζ_j a single-harmonic function. No two-wave or slowly varying envelope assumption has entered Eq. (26).

Let ℓ be the slab-normal Bloch exponent. Rotation of the carrier phase gives

$$\alpha = p\cos\Psi - \nu\sin\Psi, \qquad \ell = p\sin\Psi + \nu\cos\Psi,$$
$$p = \alpha\cos\Psi + \ell\sin\Psi, \qquad \nu = -\alpha\sin\Psi + \ell\cos\Psi. \tag{27}$$

Consequently each harmonic has the surface phase

$$e^{-\mathrm{i}p\tau_f - \mathrm{i}qz - \mathrm{i}(\nu - mK)\xi} = e^{-\mathrm{i}\alpha_m x' - \mathrm{i}qz - \mathrm{i}(\ell - mK_y)y}. \tag{28}$$

The incident wave fixes (α, q), not one common p for all internal modes. Different ℓ in Eq. (27) generally require different p and different fringe polarisation vectors. This is why one scalar Bremmer sector, or one global pair of fringe polarisations, cannot alone satisfy the finite conical plate's boundary conditions.

The exact identities $f_m = R_m - S_m$ and

$$(\nu - mK)f_m = \sum_h k_h(R_{m-h} + S_{m-h}) \tag{29}$$

allow reconstruction by Eq. (17). Summing these bulk sectors at fixed (α, q) gives the full surface Fourier fields. The next step changes their directional basis; it does not discard either scalar family.

C. Exact change to surface-directed electromagnetic amplitudes

Write each physical tangential Fourier coefficient as $e^{\mathrm{i}mK_y y}F_m(y)$, so a bulk mode has $F_m \propto e^{-\mathrm{i}\ell y}$. In the original (x', y, z) axes take

$$F_m = (E_{x,m}, E_{z,m}, g_{x,m}, g_{z,m})^{\mathsf{T}}. \tag{30}$$

The y components of the curl equations eliminate the normal fields:

$$g_{y,m} = (qE_{x,m} - \alpha_m E_{z,m})/k_0,$$
$$E_{y,m} = \frac{1}{k_0}\sum_n a_{m-n}(\alpha_n g_{z,n} - q g_{x,n}). \tag{31}$$

The other four components then give, with derivatives now in y,

$$\mathrm{i}E'_{x,m} = mK_y E_{x,m} + \alpha_m E_{y,m} - k_0 g_{z,m},$$
$$\mathrm{i}E'_{z,m} = mK_y E_{z,m} + qE_{y,m} + k_0 g_{x,m},$$
$$\mathrm{i}g'_{x,m} = mK_y g_{x,m} + \alpha_m g_{y,m} + k_0\sum_n \varepsilon_{m-n}E_{z,n},$$
$$\mathrm{i}g'_{z,m} = mK_y g_{z,m} + qg_{y,m} - k_0\sum_n \varepsilon_{m-n}E_{x,n}. \tag{32}$$

Equations (31)–(32) follow from the same fields reconstructed from PSM, before any Fourier truncation. The proper rotation of Sec. 2 turns them into Eq. (10). Thus the relation to RCWA is an equality of the complete Maxwell representations, not an assumption that a four-wave truncation is rigorous.

For $i = (m, \sigma_i, \rho_i)$ use the background electric vectors e_i specified in Appendix B, and set

$$\begin{aligned} \widehat{k}_i &= (\alpha_m, \sigma_i\gamma_m, q)/\beta, \\ h_i &= n_0 \widehat{k}_i \times e_i, \qquad w_i = \frac{1}{r_m}(e_{ix}, e_{iz}, h_{ix}, h_{iz})^{\mathsf{T}}, \\ r_m &= \sqrt{\gamma_m}, \qquad \widetilde{w}_i^{\mathsf{T}} = \frac{\sigma_i k_0}{2r_m}(-h_{iz}, h_{ix}, e_{iz}, -e_{ix}). \end{aligned} \tag{33}$$

For the four states of one order, $\widetilde{w}_i^{\mathsf{T}} w_j = \delta_{ij}$. This is a bilinear reciprocal-flux dual, without complex conjugation; it continues to closed orders. It follows directly from $e_i \cdot e_i = 1$, $e_i \cdot \widehat{k}_i = 0$ and the two opposite signs of normal propagation.

Define the surface directional amplitudes by

$$F_m = \sqrt{\gamma_0} \sum_{i:\, m_i = m} w_i a_i, \qquad a_i = \widetilde{w}_i^{\mathsf{T}} F_m / \sqrt{\gamma_0}. \tag{34}$$

They are amplitudes of the *tangential* state. Inside the modulated grating its normal electric field must still be reconstructed by Eq. (31). In either uniform exterior the full electric amplitude of state i is $\sqrt{\gamma_0/\gamma_m}\, a_i e_i$. Its open-order power relative to a unit incident electric wave is $|a_i|^2$. This explains the flux convention used in the efficiency formulae.

D. Derivation of the conical coupling vertices

At zero modulation Eq. (32) diagonalises as $\mathrm{i}a_i' = \lambda_i a_i$, with $\lambda_i = mK_y + \sigma_i\gamma_m$. Let $V_T = \varepsilon/n_0^2 - 1$ and $V_N = 1 - n_0^2/\varepsilon$, and use their Fourier coefficients in the grating phase convention. Projecting the material perturbation with the dual in Eq. (33) gives the exact infinite-order directional system

$$\boxed{\mathrm{i}a_i' = \lambda_i a_i + \sum_j \mathcal{V}_{ij} a_j,} \tag{35}$$

$$\mathcal{V}_{ij} = \frac{\sigma_i \beta^2}{2 r_m r_n}\left[(V_T)_{m-n} T_{ij} + (V_N)_{m-n} V_{ij}\right], \tag{36}$$

$$T_{ij} = e_{ix}e_{jx} + e_{iz}e_{jz}, \qquad V_{ij} = e_{iy}e_{jy}. \tag{37}$$

Here V_{ij} is an overlap, distinct from the calligraphic coupling $\mathcal{V}_{ij}$. To see the two material factors explicitly, the perturbation applied to the unnormalised background vector of state j has components

$$\delta G_{mn} t_j = \begin{pmatrix} \alpha_m n_0^2\, \delta a_{m-n} e_{jy} \\ q n_0^2\, \delta a_{m-n} e_{jy} \\ k_0 \delta\varepsilon_{m-n} e_{jz} \\ -k_0 \delta\varepsilon_{m-n} e_{jx} \end{pmatrix}, \tag{38}$$

where $t_j = r_n w_j$, $\delta a_h = a_h - \delta_{h0}/n_0^2$ and $\delta\varepsilon_h = \varepsilon_h - n_0^2\delta_{h0}$. The dual's last two entries give $(\delta\varepsilon/n_0^2)T_{ij}$. Its first two use $\alpha_m h_{iz} - q h_{ix} = k_0 n_0^2 e_{iy}$ and give $(-n_0^2\delta a)V_{ij}$. These are precisely V_T and V_N in Eq. (36). The reciprocal-permittivity contribution is therefore required by normal-field elimination; it is not an empirical TM correction.

Put $c = \cos(K\xi + \phi_g)$. For $|\mu| < 1$,

$$\begin{aligned} V_T &= 2\mu c + \mu^2 c^2, \\ V_N &= 1 - (1+\mu c)^{-2} = 2\mu c - 3\mu^2 c^2 + 4\mu^3 c^3 + O(\mu^4). \end{aligned} \tag{39}$$

Substituting the Fourier coefficients of c, c^2, c^3 into Eq. (36) yields

$$\mathcal{V}_{ij} = \sum_{r=1}^{3} \mu^r c_{ij}^{(r)} + O(\mu^4), \qquad (A_r, B_r) = (2,2), (1,-3), (0,4). \tag{40}$$

This derives the vertex table in Appendix B, including the otherwise unexplained -3 and 4. The off-diagonal polarisation terms contain

$$T_{sp} = \frac{\sigma_j \gamma_n q(\alpha_n - \alpha_m)}{\beta b_m b_n}, \quad T_{ps} = \frac{\sigma_i \gamma_m q(\alpha_m - \alpha_n)}{\beta b_m b_n}. \tag{41}$$

Thus conical s–p mixing appears explicitly when $q \neq 0$. At $q = 0$ these terms vanish, and the surface system separates into the altitude-only TE and TM sectors, up to harmless signs of their basis vectors.

E. Driven companions and the canonical bulk reduction

Choose the four principal states $\mathcal{P}$ in Appendix B. They must remain coupled because $\lambda_{0,+} - \lambda_{1,\tau}$ is small near Bragg matching. Write the perturbation blocks as

$$c^{(r)} = \begin{pmatrix} A_r & B_r \\ C_r & D_r \end{pmatrix}_{\mathcal{P},\mathcal{Q}}, \qquad L_0 = \mathrm{diag}(\Lambda_P, \Lambda_Q). \tag{42}$$

In this subsection A_r, B_r, C_r, D_r are matrices, not the scalar vertex weights. For a principal unperturbed component $a_i \propto e^{-\mathrm{i}\lambda_i y}$, the first-order driven companion satisfies

$$\mathrm{i}a_v' = \lambda_v a_v + \mu c_{vi}^{(1)} a_i. \tag{43}$$

Its particular response is $\mu X_{vi} a_i$, where

$$X_{vi} = \frac{c_{vi}^{(1)}}{\lambda_i - \lambda_v}. \tag{44}$$

There is also a free companion wave; the faces determine its coefficient. Discarding this homogeneous part would miss the face correction derived below.

For a systematic, symmetrically normalised reduction, change coordinates by

$$\begin{aligned} a &= e^S z, \qquad z = (z_P, z_Q)^{\mathsf{T}}, \\ S &= \mu S_1 + \mu^2 S_2 + O(\mu^3), \\ S_1 &= \begin{pmatrix} 0 & Y \\ X & 0 \end{pmatrix}, \qquad S_2 = \begin{pmatrix} 0 & Y^{[2]} \\ X^{[2]} & 0 \end{pmatrix}. \end{aligned} \tag{45}$$

Here S is a block transformation matrix, distinct from the scalar PSM amplitude used earlier. All blocks and gaps are independent of y after the Floquet gauge. For $L = L_0 + \sum_{r\geq 1} \mu^r c^{(r)}$, the transformed generator is

$$e^{-S} L e^{S} = L + [L, S] + \tfrac{1}{2}[[L, S], S] + \cdots. \tag{46}$$

Cancelling its off-block terms at orders μ and μ^2 gives

$$\begin{aligned} [L_0, S_1] &= -c_{\mathrm{off}}^{(1)}, \\ [L_0, S_2] &= -c_{\mathrm{off}}^{(2)} - [c_{\mathrm{diag}}^{(1)}, S_1]. \end{aligned} \tag{47}$$

With $d_{iv} = \lambda_i - \lambda_v$, these equations yield

$$\begin{aligned} Y_{iv} &= -c^{(1)}_{iv}/d_{iv}, \\ X^{[2]}_{vi} &= (C_2 + D_1 X - X A_1)_{vi}/d_{iv}, \\ Y^{[2]}_{iv} &= (-B_2 - A_1 Y + Y D_1)_{iv}/d_{iv}. \end{aligned} \tag{48}$$

Taking the principal diagonal block of the same commutator expansion gives

$$\begin{aligned} H^{(0)} &= \Lambda_P, \qquad H^{(1)} = A_1, \\ H^{(2)} &= A_2 + \tfrac{1}{2}(B_1 X - Y C_1), \\ H^{(3)} &= A_3 + \tfrac{1}{2}(B_1 X^{[2]} + B_2 X - Y^{[2]} C_1 - Y C_2). \end{aligned} \tag{49}$$

The factor $1/2$ and the two endpoint denominators in the appendix therefore follow from the canonical change of coordinates. They are not obtained by inserting an averaged detuning into a two-wave formula. A further off-block S_3 cancels the cubic off-block residual; $[L_0, S_3]_{PP} = 0$, so it does not alter Eq. (49).

At this order the required intermediate diffraction orders are only $-1, 0, 1, 2$: a product contributing through cubic order cannot leave this set and return to a principal endpoint in orders 0 or 1. Both directions and both polarisations of each companion order are retained. Remote orders remain in the exact infinite system and in the error analysis; they are not asserted to vanish physically.

The reduced propagation law is

$$\boxed{\mathrm{i} z_P' = H z_P, \qquad H = \Lambda_P + \mu A_1 + \mu^2 H^{(2)} + \mu^3 H^{(3)}.} \tag{50}$$

For separated principal–companion branches, the first omitted formal bulk coefficient is fourth order. The resonant principal exchange is kept unexpanded in $U = e^{-\mathrm{i}Hd}$. Appendix B reduces this exponential to a quartic closed expression. Neither formal order counting nor the existence of the exponential supplies an error certificate by itself.

F. Deriving the face terms from the free companion waves

Expansion of the *same* coordinate map, Eq. (45), gives

$$\begin{aligned} a_P &= (\mathsf{I} + \tfrac{1}{2}\mu^2 Y X) z_P + \mu Y z_Q + O(\mu^3), \\ a_Q &= \mu X z_P + z_Q + O(\mu^2), \end{aligned} \tag{51}$$

where the first remainder uses $z_Q = O(\mu)$ for the specified illumination. The factor $\frac{1}{2}YX$ is the PP block of $S_1^2/2$. Set $z_P(0) = c_0$, $z_P(d) = U c_0$. No companion wave is incident from either exterior, so

$$\begin{aligned} &\Pi_+ a_Q(0) = 0, && \Pi_- a_Q(d) = 0, \\ &\Pi_+ z_Q(0) = -\mu \Pi_+ X c_0 + O(\mu^2), \\ &\Pi_- z_Q(d) = -\mu \Pi_- X U c_0 + O(\mu^2). \end{aligned} \tag{52}$$

For closed orders these are the same conditions that remove fields growing away from the slab. Let E^+ propagate the positive companions from 0 to d, and E^- propagate the negative companions from d to 0. Their diagonal leading entries are given in the appendix. Thus

$$\begin{aligned} z_Q(0) &= -\mu(\Pi_+ X + E^- X U) c_0 + O(\mu^2), \\ z_Q(d) &= -\mu(E^+ X + \Pi_- X U) c_0 + O(\mu^2). \end{aligned} \tag{53}$$

Substitution into Eq. (51) produces

$$\begin{aligned} &a_P(0) = F_0 c_0, && a_P(d) = F_d c_0, \\ &A_f = \mathsf{I} + \tfrac{1}{2}\mu^2 Y X. \end{aligned} \tag{54}$$

$$\boxed{\begin{aligned} F_0 &= A_f - \mu^2 Y(\Pi_+ X + E^- X U), \\ F_d &= A_f U - \mu^2 Y(E^+ X + \Pi_- X U). \end{aligned}} \tag{55}$$

These are the finite-plate launch and extraction corrections, including the free waves required to satisfy both faces.

For transmission the negative-going orders 0 and 1 can form their own nearly resonant quartet, particularly near the unslanted limit. Propagating them separately is then inappropriate over a conversion length. The implemented model uses $E^-|_{\mathcal{B}} = e^{+\mathrm{i}H_b d}$, with the quadratic canonical generator H_b derived by the same reduction on $\mathcal{B}$ and specified in the appendix. It is a resummation within the face approximation. The reflection model tested here keeps the other companion propagation diagonal; resonances requiring additional coupled companion blocks lie outside that simplification. The quoted formal face order is not a uniform theorem for arbitrary thickness or vanishing spectral gaps.

Let $E_0 = (\mathsf{I}_2, 0)^{\mathsf{T}}$ inject the incident Jones vector and let $P_1 = (0, \mathsf{I}_2)$ select the order-1 principal pair. Transmission imposes $F_0 c_0 = E_0 b$, while reflection imposes the incident order-0 pair at $y = 0$ and zero incident order-1 pair at $y = d$. Hence

$$\begin{aligned} J_T &= e^{\mathrm{i}K_y d} P_1 F_d F_0^{-1} E_0, \\ J_R &= P_1 F_0 B_R^{-1} E_0, \qquad B_R = \begin{pmatrix} (\mathsf{I}_2, 0) F_0 \\ P_1 F_d \end{pmatrix}. \end{aligned} \tag{56}$$

The transmission phase restores Eq. (28). The efficiencies are $\eta_{T,R}(b) = \|J_{T,R} b\|_2^2$. Expanding these fixed 4×4 inverses by cofactors gives the boxed scalar formulae in Appendix B.

G. What is exact, and what is approximated

The chain from locally refracted PSM to the scalar equations, infinite Floquet system and exact surface vertices is a change of representation of Maxwell's equations. The approximation enters when the material vertices and canonical bulk generator are retained through μ^3, the physical face map through μ^2, and companion propagation is simplified as stated above. No fit to RCWA determines these coefficients. At $q = 0$ the vanishing mixed vertices separate every step into the previous TE and TM constructions. At $q \neq 0$ the four principal amplitudes generally remain coupled, so an angle substitution in two independent altitude-only formulae is insufficient.

4. CLOSED FORMS AND COMPARISON METRIC

A. Approximation being tested

The corrected model keeps principal states $(0, +, s)$, $(0, +, p)$, $(1, \tau, s)$ and $(1, \tau, p)$. Its bulk generator is

$$H = H^{(0)} + \mu H^{(1)} + \mu^2 H^{(2)} + \mu^3 H^{(3)}, \tag{57}$$

and its face reconstruction is retained through μ^2, as derived in Sec. 3. Appendix B collects all coefficients and the closed evaluation formulae. Thickness propagation remains unexpanded.

The leading comparison retains only the two principal-wave couplings. In the scattering-plane channels $\rho = \sigma, \pi$, put

$$g_\sigma = \frac{k_0^2 n_0 n_1}{2\sqrt{\gamma_0\gamma_1}}, \qquad g_\pi = C_\tau g_\sigma, \tag{58}$$

$$C_\tau = \frac{\alpha(\alpha + K_x) + \tau\gamma_0\gamma_1 + q^2}{\beta^2}, \quad \delta_\tau = \frac{\gamma_0 - K_y - \tau\gamma_1}{2}. \tag{59}$$

The elementary efficiencies are

$$\eta^{(1)}_{T,\rho} = g_\rho^2 \frac{\sin^2(d\sqrt{\delta_+^2 + g_\rho^2})}{\delta_+^2 + g_\rho^2}, \tag{60}$$

$$\eta^{(1)}_{R,\rho} = \left| \frac{g_\rho \sinh(w_\rho d)/w_\rho}{\cosh(w_\rho d) + \mathrm{i}\delta_- \sinh(w_\rho d)/w_\rho} \right|^2, \tag{61}$$

$$w_\rho^2 = g_\rho^2 - \delta_-^2. \tag{62}$$

Their complex amplitudes are rotated into the same exterior s, p bases before comparison. In conical geometry the corrected same-order blocks can mix these channels, so replacing g_σ, g_π by two corrected scalar couplings is not the model tested.

B. An error uniform over input polarisation

For any two order-1 Jones matrices C and J, define

$$D = C^\dagger C - J^\dagger J. \tag{63}$$

The exact worst absolute efficiency error over all input polarisations is

$$e(C, J) = \max_{\|b\|_2 = 1} |b^\dagger D b| = \|D\|_2 . \tag{64}$$

For this Hermitian 2×2 matrix, no optimisation or angular sampling is needed:

$$e(C, J) = \left| \frac{D_{11} + D_{22}}{2} \right| + \sqrt{\left(\frac{D_{11} - D_{22}}{2} \right)^2 + |D_{12}|^2}. \tag{65}$$

The diagonal entries are real. This is an absolute efficiency difference: 10^{-4} corresponds to 0.01 percentage point. Relative errors at diffraction zeros are deliberately not used. Complex Jones errors and individual polarisation-conversion power errors are also retained in the data archive.

5. NUMERICAL BENCHMARK DESIGN AND VALIDATION

A. Declared 504-point sweep

All sweep cases use $\lambda = 0.532\,\mu$m, $n_0 = 1.5$, $\phi_g = 0$ and the sinusoidal-index profile of Eq. (1). The three families are listed in Table 1. Each family is combined with

$$\begin{aligned} &\varphi \in \{120, 135, 150, 165\}^\circ, \\ &n_1 \in \{0.003, 0.006, 0.012, 0.024, 0.048, 0.096\}, \\ &u \in \{-2, -1, -0.35, 0, 0.35, 1, 2\}, \\ &\theta = \theta_0 + u(n_1/0.02)^\circ, \qquad d = 0.16/n_1\ \mu\text{m}. \end{aligned} \tag{66}$$

For each nominal direction, the period is fixed by bare first-order Bragg matching,

$$K = 2\beta[\cos\theta_0 \cos\Psi - \sin\theta_0 \cos\varphi \sin\Psi]. \tag{67}$$

It is not retuned after changing u. This keeps μd fixed while testing detuning on a modulation-dependent scale. The sweep intentionally includes stress cases with additional open orders.

Table 1. Families used in the sweep and certificates. Angles are in degrees. The periods shown apply at $\varphi = 150^\circ$ and $u = 0$.

Family	τ	Ψ	θ_0	Λ (μm)
Slanted transmission	$+1$	80	25	0.342464
Slanted reflection	-1	30	20	0.184358
Unslanted transmission	$+1$	90	25	0.484520

There are 103 such points and 401 with both neighbours $-1, 2$ closed. No attempted grid points were discarded.

Every grid point is calculated with $N = 7, 11, 15, 19$ orders, using the original reciprocal-coefficient factorisation. The $N = 19$ electric Galerkin result is also evaluated. The largest $15 \to 19$ change in the polarisation-uniform efficiency is 3.48×10^{-11}; the largest $N = 19$ difference between the two factorisations is 1.36×10^{-12}. These are convergence observations, not infinite-tail certificates.

The largest all-order power-balance residual is 5.83×10^{-11}. Zero modulation and zero thickness give zero order-1 amplitude. In-plane cross-polarisation amplitudes are below 2.1×10^{-16}. Phase-shift and azimuth-mirror power checks agree within 8.2×10^{-13}; changing the numerical short-step length by a factor of four changes the Jones matrix by at most 2.8×10^{-12} in the control cases.

B. Connection with the published conical example

Figure 1 recomputes the geometry of Fig. 5 in Ref. [1]: $\lambda = 0.500\,\mu$m, $\Lambda = 1/3\,\mu$m, $\Psi = 90^\circ$, $\delta = 30^\circ$, and $\theta = \arcsin(1/\sqrt{3}) = 35.2644^\circ$. Here the material is a sinusoidal *permittivity*, $\varepsilon = 2.25 + 0.01\cos\chi$, as in that example. Its reciprocal coefficients are $(-r)^{|h|}/\sqrt{2.25^2 - 0.01^2}$, with $r = 0.01/[2.25 + \sqrt{2.25^2 - 0.01^2}]$.

The calculation reproduces the reported polarisation mixing and the near-unity conversion for incident $E \perp K$ or $H \perp K$. On a 1 μm thickness grid, the first maxima occur near 61 and 122 μm, respectively, with efficiencies above 0.99996. This is a reproduction of the published geometry and qualitative curve structure, not a precision comparison to digitised original data. The original sinusoidal-permittivity example is kept separate from the sinusoidal-index F-PSM benchmark.

C. Observed accuracy and its limits

Figure 2 shows maxima over azimuth and detuning at each modulation. The corrected model has smaller error than the leading model at every sampled point. The sweep maxima are summarised in Table 2. Over all 504 cases the worst corrected error is 1.5264×10^{-2}, versus 1.0651×10^{-1} for the leading model. The worst corrected point is unslanted transmission at $\varphi = 120^\circ$, $n_1 = 0.096$, $u = 2$, with additional open orders. It is a stress case, not evidence of uniform high accuracy for arbitrary conical geometry.

In the 268-point subset with $\mu \le 0.016$ and both neighbours closed, the largest corrected error is 4.029×10^{-5}, versus 1.990×10^{-3} for the leading result. Including additional open orders raises the weak-modulation maximum to 2.745×10^{-4}. Thus small index modulation by itself is an inadequate applicability criterion.

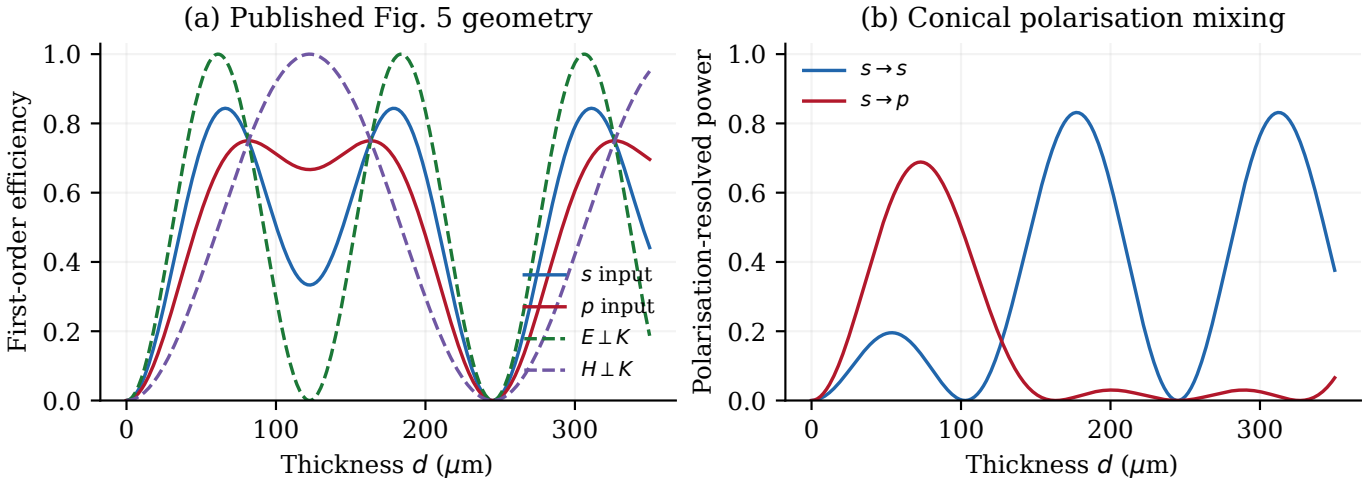


Fig. 1. Independent RCWA calculation in the geometry of Moharam and Gaylord's Fig. 5. (a) Total order-1 transmission for four incident polarisations. (b) Resolved output powers for s input show the conical $s \to p$ conversion. These are newly calculated curves, not extracted data from the original figure.

Table 2. Observed maximum absolute efficiency errors over azimuth and detuning. Each row represents 28 points. L is leading order and C is the cubic-bulk/quadratic-face result. These maxima are sampled results, not bounds over continuous parameter ranges.

	$\mu = 0.002$	$\mu = 0.016$	$\mu = 0.064$
Slanted T, L	$2.205\,10^{-4}$	$1.990\,10^{-3}$	$1.407\,10^{-2}$
Slanted T, C	$1.966\,10^{-7}$	$4.029\,10^{-5}$	$6.289\,10^{-4}$
Slanted R, L	$1.032\,10^{-4}$	$8.640\,10^{-4}$	$6.365\,10^{-3}$
Slanted R, C	$2.714\,10^{-9}$	$1.658\,10^{-6}$	$1.378\,10^{-3}$
Unslanted T, L	$1.140\,10^{-4}$	$5.308\,10^{-3}$	$1.066\,10^{-1}$
Unslanted T, C	$4.771\,10^{-7}$	$2.745\,10^{-4}$	$1.527\,10^{-2}$

The response diagnostic $|\mu| \max(|X_{vi}|, |Y_{iv}|)$ is useful but is not an error certificate (Fig. 3). In the worst stress case it is approximately 0.188. Gap size, detuning, thickness, omitted companion propagation and proximity to Rayleigh thresholds also matter. In a uniformly separated regime, the formal neglected bulk term is $O(\mu^4)$ and the neglected face term is $O(\mu^3)$. With $d = O(\mu^{-1})$, one expects a bulk accumulation of order μ^3, but this order count supplies neither a numerical constant nor a uniform bound near small gaps.

6. VALIDATED FINITE-RCWA ERROR CERTIFICATES

All physical inputs for a certificate are specified by exact decimal strings and integer-degree angles. Irrational quantities, including π, trigonometric functions, roots, exponentials and linear solves, are enclosed with Arb complex ball arithmetic through python-flint [8, 9]. No ordinary floating-point result is promoted to an exact physical input. Operations returning nonfinite enclosures are rejected.

The finite RCWA calculation is enclosed using the same scattering composition as above. The analytic closed form is enclosed by evaluating its 4×4 matrix exponential and boundary solve in ball arithmetic. These are analytic identities for the radical/adjugate expression, not a change of model. This avoids making an interval enclosure depend on a fragile ordering of quartic roots. The delivered ordinary radical evaluator agrees with the ball-enclosed matrix function to 1.10×10^{-13} in the tested Jones entries.

Evaluating Eq. (65) in balls produces an enclosure of the exact efficiency error for the specified finite systems. A short decimal upper bound is rounded upward and then checked against the ball upper endpoint. This certifies

$$\sup_{\|b\|_2=1} |\eta_{\mathrm{CF}}(b) - \eta_{\mathrm{MG},N}(b)| \le E_N. \tag{68}$$

It includes numerical evaluation error, but it does not yet include the error between N orders and infinitely many orders.

There are 24 finite certificates: three families, $\varphi = 150°$, $n_1 \in \{0.006, 0.012, 0.024, 0.048\}$, and $u \in \{0, 1\}$, with $d = 0.16/n_1$. Each was computed at 160 and 224 bits; the Jones enclosures overlap. The maximum 160-bit Jones-ball radius is 3.89×10^{-36}. The largest certified efficiency bound is 9.774×10^{-6}, so numerical uncertainty is negligible relative to model discrepancy at these points.

Table 3 gives the on-Bragg subset. Selected $19 \to 23$ differences are below 5.42×10^{-32} in the same error metric. Figure 4 shows how ordinary double precision reaches a floor while ball calculations resolve further finite-order changes. Even a rigorously enclosed difference between two truncations is not, without an additional theorem, a bound on the remaining infinite tail.

7. A CERTIFICATE AGAINST THE FULL MAXWELL PROBLEM

A. Outgoing volume-integral operator

For the infinite-order problem, write

$$\chi = \frac{\varepsilon - n_0^2}{n_0^2} = 2\mu \cos\chi_g + \mu^2 \cos^2\chi_g, \tag{69}$$

where $\chi_g = K\xi + \phi_g$. Its nonzero Fourier magnitudes are

$$|\chi_0| = \mu^2/2, \quad |\chi_{\pm1}| = |\mu|, \quad |\chi_{\pm2}| = \mu^2/4, \tag{70}$$

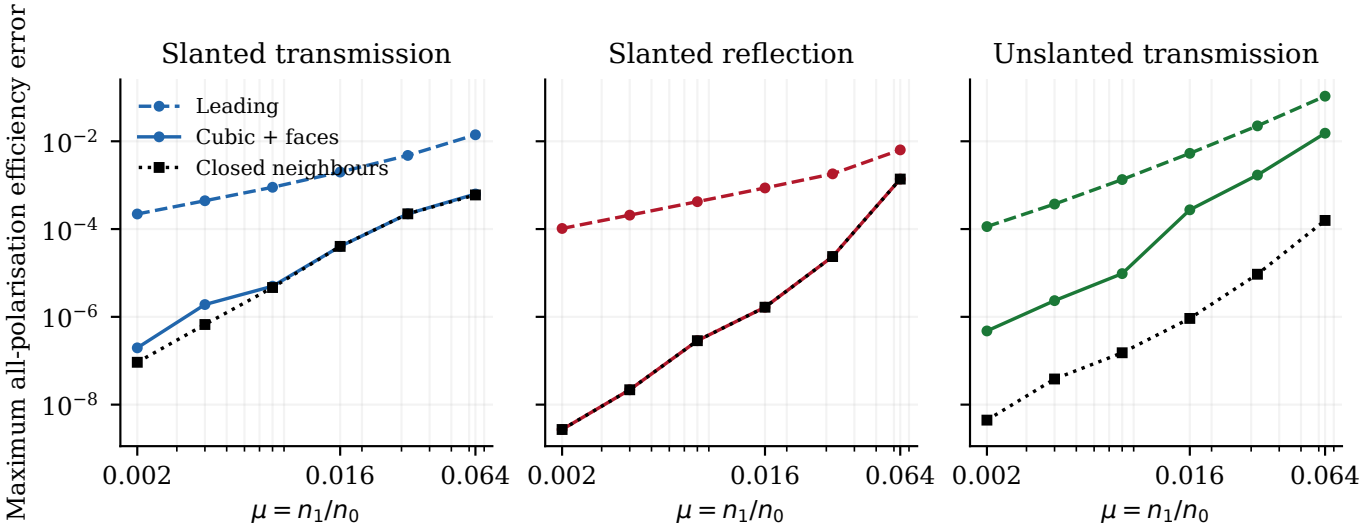


Fig. 2. Maximum polarisation-uniform efficiency error over 28 azimuth/detuning points per modulation and family. Dashed curves: leading two-wave model. Solid curves: corrected closed forms. Dotted black curves: corrected result restricted to closed neighbouring orders. The sweep keeps μd fixed; rapid face phases can prevent a smooth power-law trend.

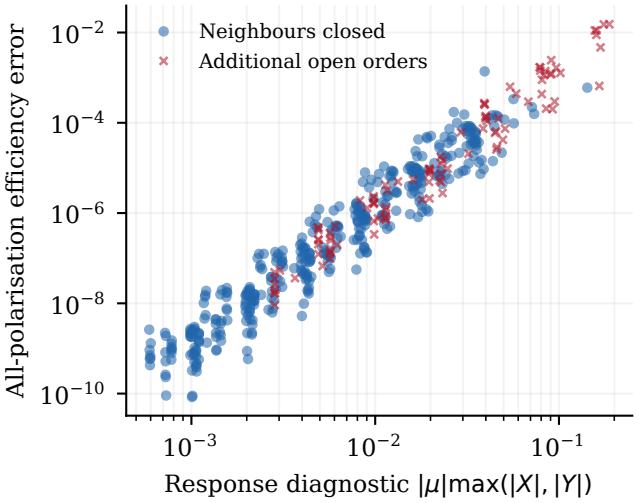


Fig. 3. All 504 corrected errors against the first-response diagnostic. Crosses identify additional open neighbouring orders. The diagnostic helps identify difficult regimes but does not uniquely predict the error.

and $c_\chi = \sum_h |\chi_h| = 2|\mu| + \mu^2$.

Let $\mathcal{E}_m(y)$ be the three-component electric amplitude for order m, or the 3×2 matrix for two incident basis polarisations. Use the Banach norm

$$\|\mathcal{E}\|_{\mathcal{X}} = \sum_{m\in\mathbb{Z}} \operatorname*{ess\,sup}_{0<y<d} \|\mathcal{E}_m(y)\|_F . \tag{71}$$

For one column, the Frobenius norm is the Euclidean norm. Bounding the two-column field in this norm also bounds its action on every unit Jones vector.

With background index n_0 on both sides, the outgoing electric volume-integral equation is

$$\mathcal{E} = \mathcal{E}^{\mathrm{inc}} + \mathcal{K}\mathcal{E}, \qquad \mathcal{K} = \beta^2 \mathcal{G}\chi. \tag{72}$$

Table 3. Upward-rounded finite-system certificates E_{19} at $\varphi = 150°$, $u = 0$, $d = 0.16/n_1$. Each applies to every normalised complex input Jones vector.

n_1	Slanted T	Slanted R	Unslanted T
0.006	$2.287\,10^{-8}$	$1.527\,10^{-8}$	$3.838\,10^{-8}$
0.012	$1.669\,10^{-7}$	$7.080\,10^{-8}$	$1.239\,10^{-7}$
0.024	$1.366\,10^{-6}$	$8.393\,10^{-7}$	$8.923\,10^{-7}$
0.048	$9.774\,10^{-6}$	$5.474\,10^{-6}$	$6.541\,10^{-6}$

In order m, the scalar Green function is

$$g_m(y, y') = \frac{e^{-\mathrm{i}\gamma_m|y-y'|}}{2\mathrm{i}\gamma_m}. \tag{73}$$

The electric dyadic is $(\mathrm{I} + \nabla_m \nabla_m/\beta^2) g_m$. Its normal–normal entry includes $-\delta(y-y')/\beta^2$. Retaining this local term is essential; dropping it would give an incorrect operator bound.

Rotate the tangential axes so one lies along the tangential wavevector of magnitude $b_m = \sqrt{\alpha_m^2 + q^2}$. The absolute-value majorant of the regular sagittal block of $\beta^2\mathcal{G}_m$ has norm at most $(|\gamma_m|^2 + b_m^2)|g_m|$; the other polarisation has bound $\beta^2|g_m|$. Including the local term gives

$$\left\|\beta^2 \mathcal{G}_m\right\|_{\infty\to\infty} \le C_m, \tag{74}$$

where

$$C_m = \begin{cases} 1 + \beta^2 d/(2\gamma_m), & \gamma_m > 0, \\ 1 + (\beta^2 + 2\kappa_m^2)\dfrac{1 - e^{-\kappa_m d/2}}{\kappa_m^2}, & \gamma_m = -\mathrm{i}\kappa_m. \end{cases} \tag{75}$$

Consequently,

$$\|\mathcal{K}\|_{\mathcal{X}\to\mathcal{X}} \le \rho := c_\chi C_*, \quad C_* \ge \sup_m C_m. \tag{76}$$

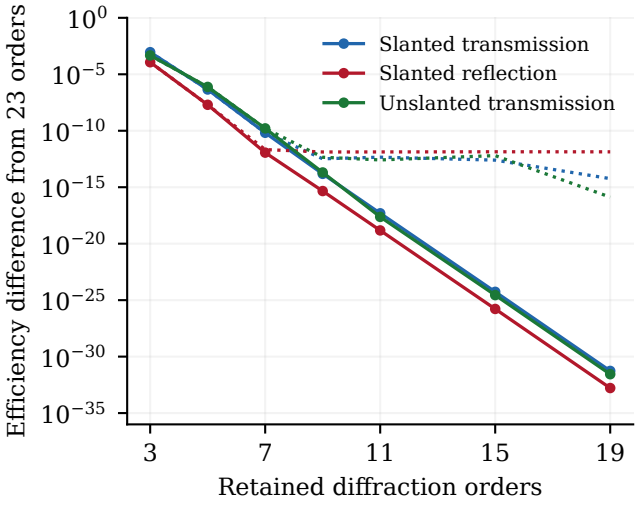


Fig. 4. Finite-order convergence at $n_1 = 0.048$, $u = 0$, $\varphi = 150°$, relative to 23-order RCWA. Solid curves are 224-bit ball upper bounds on the difference; dotted curves use double precision. This demonstrates a numerical precision floor, not an infinite-order tail bound.

A simple bound for all closed orders is $C_m \le 3 + \beta^2/\kappa_{\min}^2$. In the certificate code, all open orders and the smallest closed κ_m are enclosed within $-32 \le m \le 32$. The bounds at $m = \pm 33$ and monotonic growth of $|\alpha_m|$ beyond them certify that no smaller omitted κ_m exists. This is a proof of the tail coverage, not a presumed numerical cutoff.

B. Residual of the electric Galerkin field

Let P_M project onto $|m| \le M$. The finite electric Galerkin solution satisfies

$$\mathcal{E}^{(M)} = \mathcal{E}^{\mathrm{inc}} + P_M \mathcal{K} \mathcal{E}^{(M)}. \tag{77}$$

Eliminating its normal electric component gives exactly Eq. (10) with $\mathcal{A} = T_N(\varepsilon)^{-1}$. Independent quadrature of the projected volume-integral equation agrees with the finite Maxwell fields to 1.42×10^{-15} in the three short-slab checks supplied with the code. Its residual in the infinite equation is

$$\mathcal{R} = (\mathsf{I} - P_M)\mathcal{K}\mathcal{E}^{(M)}. \tag{78}$$

Because $\chi_h = 0$ for $|h| > 2$, only four orders can contribute: $m = \pm(M+1), \pm(M+2)$. If B_n encloses $\operatorname{ess\,sup}_y \left\| \mathcal{E}_n^{(M)}(y) \right\|_F$, then

$$R_* := \sum_{|m|>M} C_m \sum_{|n|\le M} |\chi_{m-n}| B_n \ \ge \|\mathcal{R}\|_{\mathcal{X}}. \tag{79}$$

Only $n = -M, -M+1, M-1, M$ are needed. The grating slant introduces depth-dependent phases in the physical Fourier coefficients, all of unit modulus, so it does not alter Eq. (79).

Proposition. If $\rho < 1$, the outgoing volume-integral equation has a unique solution in $\mathcal{X}$, and

$$\left\| \mathcal{E} - \mathcal{E}^{(M)} \right\|_{\mathcal{X}} \le \frac{R_*}{1-\rho}. \tag{80}$$

Proof. Subtract the projected equation from Eq. (72). The difference equals $\mathcal{K}$ times itself plus $\mathcal{R}$. The Neumann-series inverse of $\mathsf{I} - \mathcal{K}$ has norm at most $(1-\rho)^{-1}$. Equation (80) follows. The radiation Green function imposes the outgoing condition, and its distributional local term preserves the normal-field contribution. □

C. Certified depth supremum, not sampled fields

To enclose each B_n, the finite directional state is written as $c' = Ac$, where $A = -\mathrm{i}W^{-1}G_N W$ and W is the exterior directional basis. Its exact initial value is obtained from the validated scattering matrix. The depth interval is covered by 128 panels. At each centre, a Taylor expansion through degree 32 encloses the field throughout the panel. For half-width h, a rigorous state-remainder majorant is

$$c_p(y) = \sum_{k=0}^{p} \frac{A^k c(y_c)}{k!}(y - y_c)^k,$$

$$\left\| c(y) - c_p(y) \right\| \le \|c(y_c)\| \, \frac{x^{p+1} e^x}{(p+1)!}, \quad x = \|A\|_F h. \tag{81}$$

Each electric-component row is bounded by its row norm times this remainder, plus the explicitly enclosed polynomial terms. Taking component bounds, then their Frobenius norm and the maximum over panels, gives B_n. This covers the entire slab; maxima at discrete sample points alone would not justify Eq. (79). The calculation uses 512 bits, raised to 640 bits for the $d = 2\,\mu$m cases. A refined 256-panel, degree-40 calculation further tightens the most conservative reflection-tail check.

D. From field residual to efficiency error

For either open order-1 output, its radiation integral and flux normalisation give

$$\left\| J_\infty - J_{\mathrm{G},N} \right\|_2 \le \delta_J, \tag{82}$$

$$\delta_J = \frac{\beta^2 d}{2\sqrt{\gamma_0 \gamma_1}} \left(\max_h |\chi_h| \right) \frac{R_*}{1-\rho}. \tag{83}$$

The single-order convolution uses the maximum Fourier coefficient because the field-error norm already sums over all orders. The open-channel transverse projection has operator norm one. Thus a complete certificate is

$$\sup_{\|b\|_2 = 1} |\eta_{\mathrm{CF}}(b) - \eta_\infty(b)| \le E_\infty,$$

$$E_\infty = e(J_{\mathrm{CF}}, J_{\mathrm{G},N}) + \delta_J \left(2 \left\| J_{\mathrm{G},N} \right\|_F + \delta_J \right). \tag{84}$$

The Frobenius norm safely majorises the spectral norm. Every quantity on the right is evaluated outward. Unlike Eq. (68), this bound includes the infinite-order truncation error. It remains conditional on the stated ideal Maxwell material and geometry, not on unknown fabrication or material errors.

8. FULL MAXWELL CERTIFICATES AND CONCLUSIONS

The full certificates use the three families in Table 1, $\varphi = 150°$, $u = 0$, $n_1 \in \{0.006, 0.012, 0.024\}$ and $d \in \{0.5, 1, 2\}\,\mu$m: 27 exact parameter points. Both $N = 7$ and $N = 11$ Galerkin fields are enclosed; their efficiency difference lies within the sum of their separately computed tail bounds. Equation (84) is reported using $N = 11$. Figure 5 and Table 4 show the results.

The largest contraction bound is 0.818971, the largest full efficiency bound is 6.391377×10^{-7} and the largest separately bounded tail contribution is 5.106820×10^{-10}. The error is therefore predominantly the closed-form model discrepancy in this set, not an unresolved numerical or Fourier tail. The bounds

Table 4. Full Maxwell certificates at $d = 2\,\mu$m, $\varphi = 150^\circ$, $u = 0$. Numbers are rounded upward. The other 18 cases are supplied in the data archive.

Family	n_1	ρ upper	E_∞ upper
Slanted T	.006	.203495	$2.192\,10^{-9}$
	.012	.407801	$3.424\,10^{-8}$
	.024	.818851	$5.116\,10^{-7}$
Slanted R	.006	.203524	$1.047\,10^{-9}$
	.012	.407861	$1.651\,10^{-8}$
	.024	.818971	$2.501\,10^{-7}$
Unslanted T	.006	.164707	$2.396\,10^{-9}$
	.012	.330070	$3.870\,10^{-8}$
	.024	.662770	$6.392\,10^{-7}$

apply pointwise in geometry and modulation, but uniformly to every normalised complex incident polarisation. They are not certificates over the continuous angular intervals between the test points.

For the stronger reference cases $n_1 = 0.02$, $d = 8\,\mu$m, the same global contraction test yields approximately 2.64536, 2.64576 and 2.12578 for slanted T, slanted R and unslanted T. Since these exceed one, this proof does not close. That failure does not imply that the scattering solution or the closed forms are inaccurate; it reflects a sufficient norm condition that does not exploit resonant two-wave conversion. A proof for that regime would require a sharper resolvent or a block treatment that retains the resonant core exactly before bounding the remaining modes. The present report does not substitute a fitted convergence ratio for that missing proof.

The resulting conclusions are specific. The corrected conical F-PSM formulas improve every sampled comparison with the independently implemented Moharam system. Accuracy depends on geometry, open-order structure and separation from companion modes as well as modulation. Finite-system interval certificates make the numerical part of the comparison rigorous at selected strongly converting points. Full Maxwell certificates, including the infinite-order tail, have been obtained at 27 thinner-grating points. A uniform bound for arbitrary altitude, azimuth, thickness or modulation is not established.

DATA AND IMPLEMENTATION AVAILABILITY

The accompanying source archive contains this manuscript, the original calculation scripts, all 504 sweep rows, finite-system and full-Maxwell certificate records, audit outputs and figure sources. A derivation check independently projects the curl-equation Taylor coefficients onto the reciprocal-flux basis, checks the canonical commutator expansion through cubic order, and verifies the in-plane polarisation split. Its largest numerical identity residual is 3.18×10^{-14} in the three checked geometries; this is an algebra check, not an additional error certificate. The calculation code is independent of proprietary solvers. Dependencies are NumPy, SciPy, python-flint and Matplotlib; the ordinary closed-form evaluator itself uses only the Python standard library. Certificates record their exact inputs, numerical precision and outward-rounded bounds. The original article PDF is not redistributed in the archive; its DOI and institutional source are cited. This is a technical working report; author affiliations and journal-submission declarations have not been supplied.

A. SCATTERING COMPOSITION USED FOR RCWA

Let $W = (W_+, W_-)$ contain unit-electric s, p exterior waves, including the specified analytic continuation for closed orders. For a step $h = d/2^s$, partition

$$T = W^{-1} e^{-\mathrm{i}G_N h} W = \begin{pmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{pmatrix}. \tag{85}$$

In the convention $(b_L, a_R) = S(a_L, b_R)$, the four scattering blocks are

$$\begin{aligned} R &= -T_{22}^{-1} T_{21}, & T' &= T_{22}^{-1}, \\ T_f &= T_{11} + T_{12} R, & R' &= T_{12} T'. \end{aligned} \tag{86}$$

For slab A followed by B, write $S_A = (R_A, T'_A, T_A, R'_A)$ and similarly for B. Put

$$X = (\mathrm{I} - R_B R'_A)^{-1} R_B T_A, \tag{87}$$

$$Y = (\mathrm{I} - R_B R'_A)^{-1} T'_B. \tag{88}$$

Then

$$\begin{aligned} R_{AB} &= R_A + T'_A X, & T'_{AB} &= T'_A Y, \\ T_{AB} &= T_B (T_A + R'_A X), & R'_{AB} &= R'_B + T_B R'_A Y. \end{aligned} \tag{89}$$

Repeatedly composing a step with itself s times gives the full slab. The physical transmitted order m is multiplied by $e^{\mathrm{i}mK_y d}$ on restoring the gauge. This phase is also included when comparing complex Jones amplitudes.

B. SPECIFICATION OF THE CORRECTED ANALYTIC MODEL

A. States and elementary coefficients

Use the F-PSM coordinates of Sec. 2. The principal states are

$$\mathcal{P} = \{(0, +, s), (0, +, p), (1, \tau, s), (1, \tau, p)\}. \tag{90}$$

The companion set $\mathcal{Q}$ contains both polarisations of $(0, -)$ and $(1, -\tau)$, and both directions and polarisations of orders $-1, 2$. Thus $|\mathcal{P}| = 4$, $|\mathcal{Q}| = 12$. For state $i = (m, \sigma_i, \rho_i)$, set

$$\lambda_i = mK_y + \sigma_i \gamma_m, \qquad r_m = \sqrt{\gamma_m}. \tag{91}$$

For $b_m = \sqrt{\alpha_m^2 + q^2} > 0$, its electric vectors are

$$e_{m,\sigma,s} = (q, 0, -\alpha_m)/b_m, \tag{92}$$

$$e_{m,\sigma,p} = (\sigma \alpha_m \gamma_m, -b_m^2, \sigma q \gamma_m)/(\beta b_m). \tag{93}$$

At $b_m = 0$, choose $e_s = (0, 0, 1)$ and $e_p = e_s \times \widehat{k}$ instead. This is only a removable basis singularity.

For $i = (m, \sigma_i, \rho_i)$, $j = (n, \sigma_j, \rho_j)$ and $\ell = m - n$, the elementary vertices are

$$c_{ij}^{(r)} = \frac{\sigma_i \beta^2}{2 r_m r_n} f_\ell^{(r)} e^{\mathrm{i}\ell\phi_s} \left[A_r T_{ij} + B_r V_{ij} \right]. \tag{94}$$

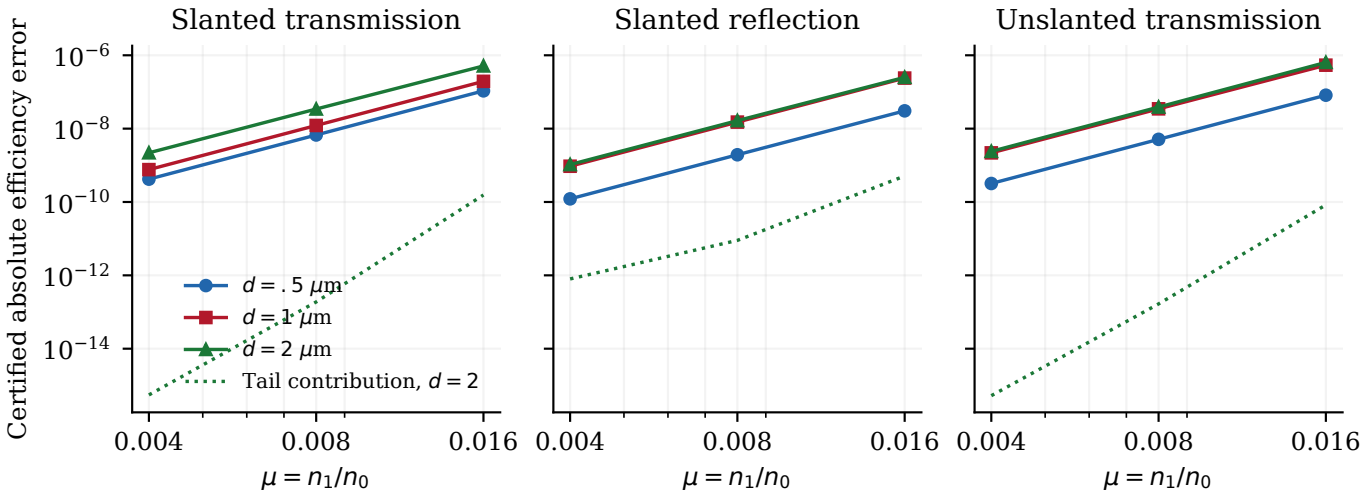


Fig. 5. Pointwise certificates against the infinite-order Maxwell problem, uniform over all incident polarisations. Solid curves connect certified points for visual guidance only. Dotted curves show the separate tail contribution at $d = 2\,\mu$m; the intervening parameter ranges have not been certified.

Here $T_{ij} = e_{ix}e_{jx} + e_{iz}e_{jz}$ and $V_{ij} = e_{iy}e_{jy}$ are bilinear overlaps without conjugation. Their explicit forms are

$$\begin{aligned} T_{ss} &= \frac{q^2 + \alpha_m\alpha_n}{b_m b_n}, & V_{ss} &= 0, \\ T_{sp} &= \frac{\sigma_j\gamma_n q(\alpha_n - \alpha_m)}{\beta b_m b_n}, & V_{sp} &= 0, \\ T_{ps} &= \frac{\sigma_i\gamma_m q(\alpha_m - \alpha_n)}{\beta b_m b_n}, & V_{ps} &= 0, \\ T_{pp} &= \frac{\sigma_i\sigma_j\gamma_m\gamma_n(\alpha_m\alpha_n + q^2)}{\beta^2 b_m b_n}, & V_{pp} &= \frac{b_m b_n}{\beta^2}. \end{aligned} \tag{95}$$

The coefficients (A_r, B_r) are $(2,2)$, $(1,-3)$ and $(0,4)$ for $r = 1,2,3$. Nonzero Fourier weights are

$$\begin{aligned} f^{(1)}_{\pm1} &= 1/2, \\ f^{(2)}_{0} &= 1/2, \quad f^{(2)}_{\pm2} = 1/4, \\ f^{(3)}_{\pm1} &= 3/8, \quad f^{(3)}_{\pm3} = 1/8. \end{aligned} \tag{96}$$

All others vanish. No powers of μ are included in $c^{(r)}$.

B. Canonical bulk reduction

For $i, j \in \mathcal{P}$ and $v, w \in \mathcal{Q}$, define

$$d_{iv} = \lambda_i - \lambda_v, \qquad X_{vi} = c^{(1)}_{vi}/d_{iv}, \quad Y_{iv} = -c^{(1)}_{iv}/d_{iv}. \tag{97}$$

The second responses are

$$X^{[2]}_{vi} = \frac{c^{(2)}_{vi} + \sum_w c^{(1)}_{vw}X_{wi} - \sum_j X_{vj}c^{(1)}_{ji}}{d_{iv}}, \tag{98}$$

$$Y^{[2]}_{iv} = \frac{-c^{(2)}_{iv} - \sum_j c^{(1)}_{ij}Y_{jv} + \sum_w Y_{iw}c^{(1)}_{wv}}{d_{iv}}. \tag{99}$$

All sums here and below are finite over the indicated sets. The corrected entries are

$$h^{(2)}_{ij} = c^{(2)}_{ij} + \frac{1}{2}\sum_v c^{(1)}_{iv}c^{(1)}_{vj}\left(d^{-1}_{iv} + d^{-1}_{jv}\right), \tag{100}$$

$$\begin{aligned} h^{(3)}_{ij} = c^{(3)}_{ij} + \frac{1}{2}\sum_v (&c^{(1)}_{iv}X^{[2]}_{vj} + c^{(2)}_{iv}X_{vj} \\ &- Y^{[2]}_{iv}c^{(1)}_{vj} - Y_{iv}c^{(2)}_{vj}), \end{aligned} \tag{101}$$

$$H_{ij} = \lambda_i\delta_{ij} + \mu c^{(1)}_{ij} + \mu^2 h^{(2)}_{ij} + \mu^3 h^{(3)}_{ij}. \tag{102}$$

Vanishing principal–companion denominators require enlargement of the core; they are not regularised by a numerical tolerance.

C. Quartic closed propagation

Let $h = \operatorname{tr} H/4$, $L = H - h\mathrm{I}$ and

$$P = -\tfrac{1}{2}\operatorname{tr} L^2, \quad Q = -\tfrac{1}{3}\operatorname{tr} L^3, \quad R = \tfrac{1}{8}(\operatorname{tr} L^2)^2 - \tfrac{1}{4}\operatorname{tr} L^4. \tag{103}$$

Let z_j be the roots of $z^4 + Pz^2 + Qz + R = 0$. For distinct roots,

$$U = e^{-\mathrm{i}Hd} = e^{-\mathrm{i}hd}\sum_{j=1}^{4} e^{-\mathrm{i}z_j d}\frac{\prod_{k\neq j}(L - z_k\mathrm{I})}{\prod_{k\neq j}(z_j - z_k)}. \tag{104}$$

The roots can be obtained without numerical eigensolving. For $Q \neq 0$, take a nonzero root S of

$$S^3 + 2PS^2 + (P^2 - 4R)S - Q^2 = 0 \tag{105}$$

by Cardano's formula, and put $A_F = \sqrt{S}$. Then

$$z_{1,2} = \tfrac{1}{2}[-A_F \pm \sqrt{-2P - S + 2Q/A_F}], \tag{106}$$

$$z_{3,4} = \tfrac{1}{2}[A_F \pm \sqrt{-2P - S - 2Q/A_F}]. \tag{107}$$

For $Q = 0$, use $z = \pm\sqrt{[-P \pm \sqrt{P^2 - 4R}]/2}$. Repeated roots use the confluent matrix-function limit; uncoupled 2×2 blocks use the elementary sine/cosine expression. The ordinary evaluator rejects unresolved nearly repeated coupled quartics, rather than reporting a spurious answer. No such rejection occurred in the benchmark grid. The validated calculation of the same function uses the rigorous matrix exponential directly.

D. Faces and boxed efficiencies

On the companion space define $\Pi_{\pm} = \mathrm{diag}(\mathbf{1}_{\sigma_v=\pm 1})$,

$$E^{+}_{vv} = \mathbf{1}_{\sigma_v=+1} e^{-\mathrm{i}\lambda_v d}, \qquad E^{-}_{vv} = \mathbf{1}_{\sigma_v=-1} e^{+\mathrm{i}\lambda_v d}. \tag{108}$$

For transmission only, the backward quartet $\mathcal{B} = \{(0,-,s),(0,-,p),(1,-,s),(1,-,p)\}$ must propagate jointly. Replace its E^{-} block by $e^{+\mathrm{i}H_b d}$, where

$$(H_b)_{vw} = \lambda_v \delta_{vw} + \mu c^{(1)}_{vw} + \mu^2 \Bigg[c^{(2)}_{vw} + \frac{1}{2} \sum_{r \notin \mathcal{B}} c^{(1)}_{vr} c^{(1)}_{rw} \left(\frac{1}{\lambda_v - \lambda_r} + \frac{1}{\lambda_w - \lambda_r} \right) \Bigg]. \tag{109}$$

The sum is over $(\mathcal{P} \cup \mathcal{Q}) \setminus \mathcal{B}$. All other entries of E^{-} remain as above. For reflection the companion propagators remain diagonal; additional resonant inter-order companion coupling lies outside that face approximation.

The face maps are

$$A_f = \mathsf{I} + \tfrac{1}{2}\mu^2 YX, \tag{110}$$

$$F_0 = A_f - \mu^2 Y(\Pi_+ X + E^{-} XU), \tag{111}$$

$$F_d = A_f U - \mu^2 Y(E^{+} X + \Pi_- XU). \tag{112}$$

Let $E_0 = (\mathsf{I}_2, 0)^{\mathsf{T}}$. For transmission set $B_T = F_0$ and let O_T be rows 3 and 4 of F_d. For reflection set B_R to rows 1 and 2 of F_0 followed by rows 3 and 4 of F_d, and let O_R be rows 3 and 4 of F_0. Then

$$J_T = e^{\mathrm{i}K_y d} O_T B_T^{-1} E_0, \qquad J_R = O_R B_R^{-1} E_0. \tag{113}$$

To make the scalar closed form explicit, for $b = T, R$ put

$$C^{b}_{\ell j} = (-1)^{\ell + j} \det(B^{b}_{\hat{\ell},\hat{j}}),$$

$$D_b = \sum_{j=1}^{4} B^{b}_{1j} C^{b}_{1j}, \qquad N^{b}_{u\ell} = \sum_{j=1}^{4} O^{b}_{uj} C^{b}_{\ell j}. \tag{114}$$

The deleted-row/column determinants are 3×3 and expand into six products. For $|b_s|^2 + |b_p|^2 = 1$,

$$\boxed{\eta_T = \frac{\sum_{u=1}^{2} |N^{T}_{u1} b_s + N^{T}_{u2} b_p|^2}{|D_T|^2}.} \tag{115}$$

$$\boxed{\eta_R = \frac{\sum_{u=1}^{2} |N^{R}_{u1} b_s + N^{R}_{u2} b_p|^2}{|D_R|^2}.} \tag{116}$$

These amplitudes already use flux normalisation, so no additional γ_1/γ_0 is applied. The material phase and output gauge phase are retained in the Jones amplitudes even though the common transmission gauge phase cancels from efficiency.

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